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TS31 | CS0025 THESIS

Material Demand, Cost Overrun and Schedule Delay Prediction for a Valenzuela Construction Firm Using LightGBM

Abstract

Construction materials are bought and used in large and uneven amounts, and a firm that runs out of cement or steel loses days while a firm that overstocks ties up money and storage. Victor M. Daet Construction and Supply, a private construction and supply firm in Valenzuela City, needs a better way to plan how much of each material to order and when. This study proposes a predictive analytics and inventory planning system that learns from the firm's own purchase, delivery and supply records, which may include the date, material, unit, quantity, unit price, supplier code and project code. The records will be obtained with the firm's written permission and anonymized before they reach the group, and they must start in 2021 or later with at least 5,000 rows, which is realistic if each purchase, delivery or supply line is one row.

However, the studies found on construction materials mostly forecast material prices or list the factors behind material problems, and none found forecasts the quantity of materials that one Philippine construction and supply firm needs or turns that forecast into an ordering decision. This study addresses that gap with the firm's own history. It forecasts demand by material category and by item and checks that the item forecasts add up to the category totals. It also uses the firm's project records to predict cost overrun and schedule delay, and tests whether material problems such as late deliveries, price changes since the quotation and stockouts help predict them. LightGBM will be the primary model, while a seasonal naive baseline, SARIMAX, XGBoost and one neural model will be the comparison models. Rainfall and typhoon season, fuel and commodity prices, the peso exchange rate and the government project calendar will be tested as possible drivers if the data can be joined lawfully. Model performance will be evaluated with rolling-origin cross-validation using the Mean Absolute Error and the Weighted Absolute Percentage Error.

The study aims to give the firm a demand forecast, a reorder guide with proposed reorder points and safety stock tested in a simulation of stockout and holding costs, and an early warning of the projects likely to run over cost or schedule. However, the topic is a working title that follows the records the firm is able to share. If the material records are not suitable, the project cost and schedule records become the main data, and the title would be revised. The results apply to this firm only.

1. Background and Rationale

Materials are a large part of what makes a construction project late or costly, and the Philippine studies keep naming them. Sumail et al. (2025) found material procurement delays among the top causes of delay and tied cost overruns to rising material costs, inflation and underestimated

cost. Bascon et al. (2023) surveyed 69 contractor responses and found inflation, cost fluctuation and material delays among the top factors, and Layno and Famadico (2024) propose a framework of overrun factors and mitigation measures for public infrastructure. Jusoh and Kasim (2017) note that materials management is an important part of construction projects, yet little research covers it. Yet a firm still has to decide, every week, how much of each material to order. A forecast built from the firm's own history can make that decision more consistent and reduce late orders and excess stock. Project outcomes matter just as much. Gondia et al. (2020) apply machine learning to the risk of delay in construction projects, and Wang et al. (2012) predict the cost and schedule success of projects with neural network ensembles and support vector machines, which leaves open whether a firm's own material records help predict those outcomes.

Forecasting methods for this kind of data are well tested. Demand for construction materials is lumpy, with many weeks of zero and a few large orders, and Croston (1972) and Syntetos and Boylan (2005) designed methods for exactly that pattern, while Teunter et al. (2011) linked such forecasts to inventory that may become obsolete. Syntetos et al. (2016) review the gap between forecasting theory and practice in supply chains, which is the gap a reorder guide has to cross. Gradient boosting (Chen & Guestrin, 2016) is a standard machine learning method for tables of features, and LightGBM is an implementation that Ke et al. (2017) report trains up to over 20 times faster than conventional boosting with almost the same accuracy. SARIMAX is a classical time series model with outside variables (Hyndman & Athanasopoulos, 2021), and Hewamalage et al. (2021) review recurrent neural networks for forecasting, so both give fair comparisons. SHAP explanations for tree models (Lundberg et al., 2020) can show which drivers move a forecast.

Research Gap

Most machine learning work on construction materials forecasts prices. Hosny et al. (2023) forecast monthly prices of steel, cement, brick, ceramic and gravel in Egypt with ARIMA and multiple regression, with errors of 1.4 to 2.8 percent. The Philippine studies found list the factors behind material problems or report survey perceptions (Sumail et al., 2025; Bascon et al., 2023). This study will contribute a demand forecast, consistent at the category and item levels, and a tested reorder guide for one Philippine construction and supply firm, while clearly keeping the results to that firm. The studies found on project overrun and delay do not use a firm's own material records to explain them. The searches found no peer-reviewed study that does either, which means not found by these searches and not that none exists.

2. Problem Statement

A construction and supply firm orders materials from experience and from the projects in hand, but its purchase, delivery and supply records are not used to forecast how much of each material will be needed or when to reorder, nor to warn which projects may run over cost or schedule. The study addresses the problem of whether LightGBM, compared with seasonal, SARIMAX, XGBoost and neural baselines, can forecast demand with useful accuracy, support a reorder guide and help predict project overrun and delay.

The study will answer the following problem: How can the purchase, delivery, supply and project records of a construction and supply firm be prepared and analyzed to forecast the demand for its materials by category and by item, to guide reorder points and safety stock and to predict which projects may overrun cost or schedule?

3. **Objectives**

General Objective

To develop and evaluate a construction material demand forecasting, inventory planning and project overrun and delay prediction system for Victor M. Daet Construction and Supply in Valenzuela City using LightGBM and comparison models.

Specific Objectives

- Obtain, clean and anonymize the firm's material purchase, delivery and supply records from 2021 onward, and profile missing, duplicated and inconsistent records.
- Analyze how demand changes with the week, the month, the rainy and dry seasons and the start of projects, and test rainfall and typhoon season, fuel and commodity prices, the peso exchange rate and the government project calendar as drivers if they can be joined lawfully.
- Forecast demand by material category and by item with LightGBM, compare it with a seasonal naive baseline, SARIMAX, XGBoost and one neural model, and check that item forecasts add up to category totals.
- Evaluate the models using rolling-origin cross-validation with MAE and WAPE, with the target that the best model's error is lower than the best baseline's by at least 10 percent (the target will be revised after the data check).
- Turn the forecasts into proposed reorder points and safety stock, and simulate stockout and holding costs if stock, lead-time and cost inputs exist.
- Develop LightGBM models for the cost overrun and the schedule delay of the firm's projects from the project records, compare them with a baseline of past project outcomes, and test whether late deliveries, price changes since the quotation and stockouts from the material records improve them.
- Explain the forecasts and the project models with SHAP, check whether the patterns stay stable over time, and deliver a dashboard of the forecast, the reorder guide and the project risk list for the firm's staff to test.

4. **Research Questions**

1. What patterns appear in the firm's material demand across weeks, seasons and project starts?
2. How accurately can demand be forecast by category and by item, do item forecasts add up to category totals, does LightGBM beat the baselines, and do the outside drivers improve the forecasts?

3. Would the proposed reorder points have reduced stockouts and holding costs in a simulation of past weeks?
4. Do late deliveries, price changes since the quotation and stockouts help predict which projects overrun their cost or schedule?

5. **Dataset and Variables**

Data item	Description
Dataset	Victor M. Daet Construction and Supply material purchase, delivery and supply records, and project cost and schedule records (private business records)
Source	The firm, with written permission [to confirm with the client]
Unit of analysis	One material line of a purchase, a delivery to a project, or a supply order; and one project for the overrun and delay outcomes
Target	Weekly quantity needed of each material, by category and by item (forecasting); cost overrun and schedule delay of each project
Predictors	Week, month, rainy or dry season, holiday indicator, material and unit, unit price, number and start dates of active projects (coded), supplier code, recent demand, days since the last order; rainfall and typhoon season, fuel and commodity prices, exchange rate and the government project calendar if available and lawful. For projects: budget, planned and actual dates, project type and size, and from the material records the number of late deliveries, price change since the quotation and stockouts
Available records	To be confirmed. The adviser's rule is at least 5,000 rows from 2021 onward, and the usable row count is unknown

Data Preparation

- Remove client, supplier, employee and project names before the data reaches the group, and replace them with codes.
- Standardize units (for example bags, kilograms and pieces), and check missing, duplicated and invalid dates and quantities.
- Aggregate the records to weekly demand by material category and by item, and mark the materials with many zero weeks as intermittent.
- Avoid leakage by splitting the data by time, using no future prices or inventory values for past predictions, and fitting all transformations inside each training period with rolling-origin cross-validation (Bergmeir & Benítez, 2012).

6. **Proposed Method**

Phase	Planned work
Data acquisition	Meet the firm, confirm which dated item, price, stock and lead-time records it keeps, and sign the data permission.
Exploration	Summarize demand by material, week and season, and check missing values, duplicates and intermittency.

Preprocessing	Standardize units, clean the records, anonymize projects and suppliers, and build calendar, lag and driver features.
Modeling	Train LightGBM as the primary model and compare it with a seasonal naive baseline, SARIMAX, XGBoost and one neural model, with item forecasts checked against category totals. Train LightGBM models for project cost overrun and schedule delay on earlier projects and test them on later projects, with and without the material features.
Evaluation	Use rolling-origin cross-validation and report MAE and WAPE (Hyndman & Koehler, 2006) on unseen periods, then simulate stockout and holding costs for the proposed reorder points.
System development	Build the dashboard with the forecast, the reorder points and safety stock, and a view for each material.
Interpretation	Use SHAP to explain what drives the forecast and check whether the patterns remain stable over time.

7. Scope, Limitations, and Ethics

Scope

- The study covers one firm, Victor M. Daet Construction and Supply in Valenzuela City, and its records from 2021 onward.
- It forecasts demand by material category and by item, proposes reorder points and safety stock for the firm's main materials, and predicts cost overrun and schedule delay for the firm's projects.
- Calendar, season and project-start variables are the main inputs. Outside data such as rainfall, fuel and commodity prices, the exchange rate and the government project calendar are used only if they are available, lawful to use and allowed by the adviser.

Limitations

- The results apply to this firm and may not hold for other firms.
- Accuracy depends on how complete and consistent the firm's records are, and the inventory simulation needs stock, lead-time and cost records that may not exist.
- The project models may have wide ranges if the firm has few finished projects, and they need project budgets and dates that may not be kept in a usable form.
- The study does not cover pricing, supplier selection, site storage, automatic ordering or construction quality.
- A firm that keeps only paper delivery receipts or yearly totals would not meet the data requirement unless the records are typed in, and the topic may then change to project cost or schedule data.
- The firm may also take government contracts and was reached through a family contact, so whether it counts as a private client is for the adviser to decide.

Ethical and Privacy Considerations

- Follow the Data Privacy Act of 2012 (Republic Act No. 10173): client, supplier, employee and project names are removed before the data reaches the group.
- Use the records only with the written permission of an authorized firm representative for the named records and research use, keep raw records outside the repository, and wait for IREC approval before collection.
- No human participants, no web scraping and no open-source data are involved.
- Present results without naming clients or suppliers, and as forecasts for planning, not as guarantees.

8. Significance of the Study

The study will demonstrate a complete data science workflow on a private firm's records: data acquisition, cleaning, exploration, forecasting at two levels, evaluation and a tested ordering guide. It may help the firm order materials more consistently and with less excess stock, and it may help other small construction and supply firms and students who want a local case study. The project has a cautious connection to SDG 12.2 through the efficient use of resources and to SDG 9.1 through reliable infrastructure; it will not directly measure progress toward either goal. [The group should confirm the SDG with the adviser.]

9. Expected Output

- A documented and anonymized analysis dataset.
- Exploratory visualizations of material demand patterns.
- A comparison of forecasting models with MAE and WAPE, at the category and item levels.
- A dashboard with the forecast, the reorder points and safety stock, and a project risk list.
- A simulation of stockout and holding costs, overrun and delay models with and without the material features, and a discussion of the limits of modeling one firm.

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Cost Overrun and Schedule Delay Prediction for Arkidom Projects Using Gradient Boosting

Abstract

Construction projects often cost more and take longer than planned, and a small firm needs to know both risks before it quotes a project and sets aside a contingency. Arkidom Development Corporation, an architecture, urban planning and construction firm in Quezon City with a small team, needs early estimates of cost overrun and schedule delay for each new project, and a contingency percentage that it can defend. This study proposes a predictive analytics system that learns from the firm's own project records, which may include project budgets, actual costs, planned and actual dates, scope, floor area, building type and materials. The records will be obtained with the firm's written permission and anonymized before they reach the group, and they must start in 2021 or later with at least 5,000 rows, which is realistic only if cost line items count as rows, so how rows are counted will be confirmed with the adviser.

However, the Philippine cost estimation studies found use only 30 to 41 public projects, and the Philippine overrun studies are surveys of perceived causes, not predictive models. This study addresses that gap with a small private firm's own history. Two linked gradient boosting models will estimate cost overrun and schedule delay, and prediction intervals will support a contingency percentage that is recommended only if the intervals prove calibrated on later projects. Material price volatility will be tested as a driver if dated prices can be matched to projects. A baseline built from past project outcomes, a simple regression model and a tree-based model will be the comparison models. The models will be trained on earlier projects and tested on later projects, with the Mean Absolute Error, the coverage and width of the intervals, and precision, recall and calibration if a yes or no overrun flag is used.

The study aims to give the firm early overrun and delay estimates with a defensible contingency range and an explanation of the main risk factors. However, with a small number of projects the ranges will be wide, and the results apply to this firm's type of projects only.

1. Background and Rationale

Flyvbjerg et al. (2003) showed that cost overruns in infrastructure projects are common and large, and the Philippine studies report the same pattern (Layno & Famadico, 2024; Sumail et al., 2025; Begay, 2024; Bascon et al., 2023). Yet a firm still has to quote a price and a schedule before a project starts. An early estimate built from the firm's own history, and a warning for projects that look like past overruns, can make quotations more consistent and reduce surprises for the firm and its clients. Overrun and delay are also linked, since a late project usually costs more, so a project team has to consider both together. Love et al. (2012) discuss optimism bias

and strategic misrepresentation as explanations for cost overruns in social infrastructure projects, which is one more reason to check an estimate against the firm's own record.

Machine learning for construction cost and delay is well studied. Hashemi et al. (2020), Elmousalami (2020) and Sadikoglu and Demirkesen (2025) review cost estimation methods, Gondia et al. (2020) apply machine learning to the risk of project delay, and Wang et al. (2012) predict cost and schedule success of projects with neural network ensembles and support vector machines. Gradient boosting (Chen & Guestrin, 2016) and random forests (Breiman, 2001) are standard methods for tables of features. Quantile regression (Koenker & Bassett, 1978) and conformal prediction (Angelopoulos & Bates, 2023) are two ways to attach a range to a prediction, and SHAP explanations for tree models (Lundberg et al., 2020) can show which risk factors drive an estimate.

Research Gap

Machine learning has been used for construction cost estimation in the Philippines on road and building projects (Roxas et al., 2019; Manalindo et al., 2024), but the samples were small and public, and the overrun studies found are surveys. The studies found estimate cost or delay separately and do not offer a calibrated contingency range for a small private firm. This study will contribute linked overrun and delay estimates with prediction intervals for one small private architecture and construction firm, while clearly keeping the results to that firm. This means not found by these searches and not that none exists.

2. Problem Statement

A small construction firm keeps project and cost records but quotes new projects from experience, sets contingency by habit, and learns about overruns and delays only after they happen. The study addresses the problem of whether gradient boosting, compared with simple baselines, can estimate cost overrun and schedule delay for a new project and support a contingency percentage with useful accuracy.

The study will answer the following problem: How can the project and cost records of a small architecture and construction firm be prepared and analyzed to estimate cost overrun and schedule delay together and to recommend a contingency range for new projects?

3. Objectives

General Objective

To develop and evaluate a cost overrun and schedule delay prediction system with contingency estimates for Arkidom Development Corporation in Quezon City using gradient boosting and baseline methods.

Specific Objectives

- Obtain, clean and anonymize the firm's project and cost records from 2021 onward, and profile missing, duplicated and inconsistent records.

- Analyze how cost overrun and schedule delay relate to the scope, floor area, building type, materials and location, and test material price volatility if dated prices can be matched to projects.
- Develop linked gradient boosting models for cost overrun and schedule delay, and compare them with a baseline of past project outcomes, a simple regression model and a tree-based model.
- Produce prediction intervals and test whether the contingency percentage they support is calibrated on later projects, using interval coverage and width.
- Evaluate the models by training on earlier projects and testing on later projects, with MAE for the numeric estimates and precision, recall and calibration if a yes or no overrun flag is used (the targets will be revised after the data check).
- Explain the risk factors behind each estimate, check whether the explanations stay stable over time, and deliver an estimator interface with the estimate, the range and the main factors for the firm's staff to test.

4. **Research Questions**

1. What relationships exist between project characteristics and cost overrun and schedule delay in the firm's records?
2. How accurately can overrun and delay be estimated for a new project, and does gradient boosting beat the baselines?
3. Are the prediction intervals calibrated enough on later projects to recommend a contingency percentage?
4. Which risk factors contribute most to the estimates, and do they stay stable over time?

5. **Dataset and Variables**

Data item	Description
Dataset	Arkidom Development Corporation project and cost records (private business records)
Source	The firm, with written permission [to confirm with the client]
Unit of analysis	One project for the overrun and delay outcomes, with cost line items as supporting records if the firm keeps them
Target	Cost overrun (actual against budget) and schedule delay (actual against planned dates), and an overrun flag if the firm defines one
Predictors	Scope, floor area, building type, materials, location, year, bill of quantities category, planned dates, and material price volatility if dated prices can be matched
Available records	To be confirmed, including the number of completed projects. The adviser's rule is at least 5,000 rows from 2021 onward

Data Preparation

- Remove client names and other identifiers before the data reaches the group.

- Check missing, duplicated and inconsistent records, and adjust costs for inflation if the records span several years.
- Avoid leakage by using only information available at the start of a project as inputs and not the final cost or the actual dates.
- Train on earlier projects and test on later projects, and keep all records of one project in the same split.

6. Proposed Method

Phase	Planned work
Data acquisition	Ask the firm for a record inventory with date range and count, agree how an overrun and a delay are defined, and sign the data permission.
Exploration	Summarize overrun and delay by building type and size, and check missing values, duplicates and outliers.
Preprocessing	Clean the records, anonymize clients, prepare categorical and numeric features, and match dated material prices to projects if possible.
Modeling	Train linked gradient boosting models for cost overrun and schedule delay and compare them with a baseline of past outcomes, a simple regression model and a tree-based model.
Evaluation	Report MAE, interval coverage and width on later projects, and precision, recall and calibration if an overrun flag is used.
System development	Build the estimator interface with the estimate, the range, the suggested contingency and the main factors.
Interpretation	Use SHAP to explain the estimates and check whether the explanations stay stable over time.

7. Scope, Limitations, and Ethics

Scope

- The study covers one firm, Arkidom Development Corporation in Quezon City, and its project records from 2021 onward.
- It estimates cost overrun and schedule delay and suggests a contingency range for the types of projects in the records.
- Inputs are project characteristics available at the start of a project.

Limitations

- A firm of this size may have few finished projects, so estimates and ranges may be wide.
- The results apply to this firm's type of projects and may not hold for other firms.
- The study does not cover design quality, structural analysis, contractor selection, procurement or site monitoring.
- The 5,000 row rule is realistic only if line items count as rows for project-level outcomes, and the adviser has to confirm that.

Ethical and Privacy Considerations

- Follow the Data Privacy Act of 2012 (Republic Act No. 10173): client names and identifiers are removed before the data reaches the group.
- Use the records only with the written permission of an authorized firm representative for the named, nonpersonal records and research use, keep raw records outside the repository, and wait for IREC approval before collection.
- No human participants, no web scraping and no open-source data are involved.
- Present results without naming clients, and as estimates, not as formal quotations.

8. **Significance of the Study**

The study will demonstrate a complete data science workflow on a small private firm's records: data acquisition, cleaning, exploration, regression, interval estimation, evaluation and a usable estimator. It may help the firm quote projects more consistently, set contingency with evidence and see overrun and delay risk early, and it may help other small firms and students who want a local case study. The project has a cautious connection to SDG 9.1 through quality and reliable infrastructure and to SDG 11 through the firm's work in Quezon City; it will not directly measure progress toward either goal. [The group should confirm the SDG with the adviser.]

9. **Expected Output**

- A documented and anonymized analysis dataset.
- Exploratory visualizations of overrun and delay patterns.
- A comparison of the models with MAE, and interval coverage and width.
- A contingency recommendation, only if the intervals are calibrated on later projects.
- An estimator interface and a discussion of the limits of modeling a small firm's projects.

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DAWN ANDREI PAMESA
TS31 | CS0025 THESIS

Demand Forecasting and Delivery Planning for Water Stations in San Juan and Marikina Using XGBoost

Abstract

Drinking water is a daily purchase, and a small water station plans its stock, staff and delivery trips from experience, which is difficult when demand changes with the season, the weekday, holidays and the way customers order. Alpha Pi Water in San Juan City and Dr. Pi Oxygenated Water in Marikina City, two stations that sell oxygenated and alkaline drinking water and share one owner, need a better basis for planning both stations together. This study proposes a predictive analytics system that uses the two stations' own sales and delivery records to forecast demand at each station and for the pair, to flag regular customers who may be lapsing, to plan delivery routes from both stations and to test whether moving stock between the stations would reduce stockouts. The records will be obtained from the owner with written permission and anonymized before they reach the group, and they must start in 2021 or later with at least 5,000 rows, which two stations make easier to reach.

However, the Philippine studies found on water businesses focus on sanitation and compliance at refilling stations, and none of the studies found forecasts demand, predicts customer retention, and plans deliveries and stock for a small group of stations. This study addresses that gap in four linked parts. Daily demand is forecast at each station and for the pair, comparing one model per station with one pooled XGBoost model trained on both, and reconciling the forecasts so that the stations add up to the total. A repeat-buying probability model is compared with an XGBoost classifier to flag lapsing customers. Each delivery area is assigned to the nearer station and routes from both stations are planned with a savings heuristic, with the trips and distance saved tested in a simulation. Finally, a stock balancing rule that moves stock between the stations is tested in a simulation of stockouts and excess stock. A seasonal naive baseline, ARIMA and Prophet will be the comparison models for the forecasts, evaluated with time-based cross-validation using Mean Absolute Error and Weighted Absolute Percentage Error.

The study aims to give the owner a forecast, a follow-up list, a delivery plan and a stock balancing guide that can be used for weekly planning. However, the results apply to these two stations, they are forecasts and not guarantees, and the business details above come from the stations' public Facebook pages and are [to confirm with the client].

1. Background and Rationale

Drinking water businesses in the Philippines operate under national sanitation rules, which include Presidential Decree No. 856 (1975) and the Department of Health's supplemental implementing rules (1999), and which rules apply to these stations is [to confirm with the client]. Yet an owner with two stations still has to decide each day how much water each station should

prepare, how many delivery trips to schedule from each and which customers to follow up. A wrong guess means empty stock and lost sales at one station while the other has stock to spare, or idle staff and wasted trips. Customers also buy without any contract, so a regular customer who quietly stops ordering is hard to notice. A forecast built from the stations' own records can organize these patterns and give the owner a consistent basis for planning.

Forecasting methods for this kind of data are well tested. Hyndman and Athanasopoulos (2021) cover the baselines and the accuracy measures, Taylor and Letham (2018) describe Prophet for data with seasonality and holidays, and Makridakis et al. (2020) compare statistical and machine learning methods on 100,000 series. Gradient boosting methods such as XGBoost (Chen & Guestrin, 2016) are standard for tables of features, and the cross-validation of time series has to respect time order (Bergmeir & Benítez, 2012). With two stations, Montero-Manso and Hyndman (2021) discuss when one model for a group of series (a global model) is better than one model for each series (a local model), and when the stations are also summed into a total, Hyndman et al. (2011) and Wickramasuriya et al. (2019) show how to reconcile forecasts of such hierarchies. For customers who buy without a contract, Buckinx and Van den Poel (2005) study partial defection of loyal customers in a non-contractual retail setting and Fader et al. (2005) model repeat buying from recency and frequency. Delivery planning from more than one station is a vehicle routing problem with multiple depots, reviewed by Montoya-Torres et al. (2015), and the savings method of Clarke and Wright (1964) gives routes from a central depot to a number of delivery points. Paterson et al. (2011) review inventory models in which stock is moved between locations (lateral transshipment), which is the idea behind balancing stock between two stations.

Research Gap

The Philippine studies about water refilling stations, the closest local businesses found, deal with sanitation, operator practices and water safety plans (Cope et al., 2013; Molina et al., 2016). The forecasting studies found, such as Makridakis et al. (2020), use large public series and not a small business's own records. This study will contribute a Philippine small-business case that links demand forecasting, a customer retention flag, delivery planning and stock balancing across two stations with one owner, while clearly keeping the results to these two stations. This means not found by these searches and not that none exists.

2. Problem Statement

An owner with two small water stations records sales and deliveries, but may not use them to plan ahead or to plan the two stations together. Without a data-driven method, stock, staff and delivery trips are planned from experience, one station can run out while the other has stock, and lapsing customers are noticed late. The study addresses the problem of whether XGBoost, compared with simple forecasting baselines, can predict demand at each station, whether one pooled model beats one model per station, whether a repeat-buying model or a classifier better identifies customers who may stop ordering, and whether the forecasts can reduce delivery trips and stockouts.

The study will answer the following problem: How can the sales and delivery records of two small water stations be prepared and analyzed to forecast demand at each station and for the pair, to identify customers who may stop ordering, to plan delivery routes from both stations and to balance stock between them?

3. Objectives

General Objective

To develop and evaluate a demand forecasting, customer retention, delivery planning and stock balancing system for Alpha Pi Water in San Juan City and Dr. Pi Oxygenated Water in Marikina City using XGBoost and baseline methods.

Specific Objectives

- Obtain, clean and anonymize the two stations' sales and delivery records from 2021 onward, and profile missing, duplicated and inconsistent records at each station.
- Analyze how demand changes with the weekday, the day of the month, holidays, the product, the order channel (for example delivery or pick-up) and the delivery area, and compare the two stations.
- Forecast daily demand at each station with XGBoost, compare one model per station with one pooled model trained on both, compare them with a seasonal naive baseline, ARIMA and Prophet, and reconcile the forecasts so that the stations add up to the total.
- Evaluate the models using time-based cross-validation with MAE and WAPE, with the target that the best model lowers WAPE by at least 10 percent against the seasonal naive baseline (the target will be revised after the data check).
- Flag lapsing customers with a repeat-buying probability model and with an XGBoost classifier, compare them on later periods, check that the flags are calibrated, and note customers who buy from both stations if they can be matched by anonymous code.
- Assign each delivery area to the nearer station, plan routes from both stations with a savings heuristic, and simulate the trips and distance saved against the current routes if trip and address records exist.
- Test a rule for moving stock between the two stations in a simulation of stockouts and excess stock if stock records exist, and deliver a dashboard with the forecasts, the follow-up list, the route plan and the stock balancing guide for the owner to test.

4. Research Questions

1. What patterns appear in demand across weekdays, months, holidays, products and channels, and how do the two stations differ?
2. How accurately can daily demand be forecast, does XGBoost beat the simple baselines, and does one pooled model beat one model per station?
3. Does a repeat-buying model or an XGBoost classifier flag lapsing customers better on later periods?

4. Would route plans from both stations and moving stock between the stations have reduced delivery trips, distance, stockouts and excess stock in a simulation of past days and weeks?

5. **Dataset and Variables**

Data item	Description
Dataset	Sales and delivery records of Alpha Pi Water (San Juan) and Dr. Pi Oxygenated Water (Marikina), one owner (private business records)
Source	The owner, with written permission [to confirm with the client]
Unit of analysis	One sale or delivery order at one station
Target	Daily demand in units sold at each station and in total (forecasting), a lapsed-customer flag (retention), trips and distance (route simulation), and stockouts and excess stock (balancing simulation)
Predictors	Station, date, weekday, day of month, holiday indicator, product, order channel, delivery area, quantity, anonymous customer code, days since last order, order frequency
Available records	To be confirmed for each station, including whether the two keep one record or two, and whether delivery areas, trips and stock are recorded. The adviser's rule is at least 5,000 rows from 2021 onward

Data Preparation

- Remove exact duplicates and check missing or invalid dates and quantities at each station.
- Replace customer names, addresses and phone numbers with anonymous codes, and keep only a delivery area for route planning.
- Put the two stations' records in one format, keep a station column, and aggregate the orders to daily demand by station, product, channel and area.
- Avoid leakage by splitting the data by time, fitting all transformations inside each training window and flagging customers only from orders before the cut-off date.

6. **Proposed Method**

Phase	Planned work
Data acquisition	Meet the owner, confirm what each station records and how orders are taken, and sign the data permission.
Exploration	Summarize demand by weekday, month, product, channel, area and station, and check missing values and duplicates.
Preprocessing	Clean the records, anonymize customers, harmonize the two stations' formats and build calendar, area and customer features.
Modeling	Train XGBoost per station and as one pooled model, compare them with a seasonal naive baseline, ARIMA and Prophet, and reconcile the station forecasts to the total. Compare a repeat-buying model with an XGBoost classifier for lapsing customers.

Evaluation	Use time-based cross-validation and report MAE and WAPE (Hyndman & Koehler, 2006) on unseen periods, the calibration of the lapsing flags, the trips and distance saved in the route simulation, and the stockouts and excess stock in the balancing simulation.
System development	Build the dashboard with the forecasts, the follow-up list, the route plan and the stock balancing guide.
Interpretation	Use feature importance to explain what drives the forecasts and the lapsing flags, and describe how the two stations differ.

7. Scope, Limitations, and Ethics

Scope

- The study covers two stations of one owner, Alpha Pi Water in San Juan City and Dr. Pi Oxygenated Water in Marikina City, and their records from 2021 onward.
- It forecasts daily and weekly demand up to four weeks ahead, flags customers at risk of lapsing, proposes next-day delivery routes from both stations and tests a stock balancing rule.
- Calendar and holiday variables are the main inputs. Outside data such as weather is used only if the owner supplies it.

Limitations

- The results apply to these two stations and may not hold for other stations.
- Accuracy depends on how complete and consistent the records are, and the route and stock simulations need delivery area, trip and stock records that may not exist.
- The stock balancing test needs an agreed assumption about how stock is moved between San Juan and Marikina and what it costs, which the owner would have to provide.
- The study does not cover water quality, product claims, sanitation compliance, pricing, automatic ordering, sensors or container tracking.
- A station that keeps only a notebook, chat messages or monthly totals would not meet the data requirement unless the records are typed in.

Ethical and Privacy Considerations

- Follow the Data Privacy Act of 2012 (Republic Act No. 10173): customer names, addresses and phone numbers are removed before the data reaches the group.
- Use the records only with the owner's written permission and keep raw records outside the repository. Collection waits for IREC approval.
- No human participants, no web scraping and no open-source data are involved.
- Present results as forecasts and plans for the owner to review, not as guarantees, and make no claim about the health effects of the product.

8. Significance of the Study

The study will demonstrate a complete data science workflow on a small Philippine business with two sites: data acquisition, cleaning, exploration, forecasting at several levels, a retention

model, routing and stock balancing simulations, evaluation and a usable dashboard. It may help the owner plan stock and deliveries across both stations and keep regular customers, and it may help other small water businesses and students who want a local case study. The project has a cautious connection to SDG 6.1 through the reliability of small drinking water suppliers and to SDG 8.3 through the growth of small enterprises; it will not directly measure progress toward either goal. [The group should confirm the SDG with the adviser.]

9. Expected Output

- A documented and anonymized analysis dataset for both stations.
- Exploratory visualizations of demand patterns and a comparison of the two stations.
- A comparison of forecasting models with MAE and WAPE, including one pooled model against one model per station, with reconciled forecasts.
- A comparison of the repeat-buying model and the classifier for lapsing customers.
- A dashboard with the forecasts, the follow-up list, the route plan and the stock balancing guide, and a discussion of the limits of two stations' data.

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DAWN ANDREI PAMESA
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Philippine Procurement Award Red Flags Using Benford's Law and Network Analysis

Abstract

The Government Procurement Reform Act (Republic Act No. 9184) requires procuring entities to publish bid opportunities and awards in the electronic procurement portal, PhilGEPS. Published awards can show patterns that deserve a second look, but a single unusual number is not evidence of wrongdoing. This study proposes a screening system for Philippine government procurement awards that combines first-digit checks based on Benford's law, date and award checks, and the network of agencies and suppliers into one transparent risk score. It uses an open compilation of PhilGEPS awards, described as MIT licensed (csiiiv/philgeps-awards-dashboard), which reports 5,481,161 awards from 2013 to 2025, of which 2,769,747 are from 2021 onward. These counts come from the group's own checks and are [UNVERIFIED] until the source link, retrieval date and counts are reproduced.

However, the studies found on procurement risk are mostly about European and Latin American data, and none found applies Benford's law and network measures together to PhilGEPS awards. Two further limits shape the design. Procurement mode and funding source are reported empty in the compilation, so explanations cannot rest on them, and the data is open and governmental, which conflicts with the adviser's rule unless a title without a private client is allowed. The study compares the combined score with a single first-digit flag and with date-only, Benford-only and network-only scores. It builds the network and score components on earlier awards, evaluates them on later awards, and tests whether the rankings change when one component is removed.

The study aims to give audit, compliance and civic groups a transparent way to choose which awards to review first. However, an indicator is a screening signal and not a finding of misconduct, no award is called fraudulent, and any check against known cases needs named, verified sources.

1. Background and Rationale

The Government Procurement Reform Act (Republic Act No. 9184, 2003) is built on transparency, competition and accountability, and it requires procuring entities to publish bid opportunities and post awards and contracts in the electronic portal. The Philippines' open contracting data commitment (Open Government Partnership, n.d.) and the Open Contracting Partnership's (2024) guide to red flags in procurement both treat published data as a way to find risk. Yet published data is only useful if someone can turn it into a short list of awards to review.

Procurement red flags are well studied outside the Philippines. Fazekas et al. (2016) built an objective corruption risk index from public procurement data, Fazekas and Kocsis (2020) used

such indicators to compare countries, and Ferwerda et al. (2017) discuss how to choose the right indicators. Decarolis and Giorgiantonio (2022) test corruption red flags on Italian calls for tenders, Gallego et al. (2021) build an early warning model of malfeasance in public procurement, and Dávid-Barrett and Fazekas (2020) study favoritism in procurement after a change of government. Schneider dos Santos et al. (2025) map the data-driven methods used to detect fraud in public procurement. Benford's law describes how often each first digit appears in many natural data sets. Newcomb (1881) noticed it, Hill (1995) gave a statistical derivation, and Fewster (2009) gives a simple explanation, while Goodman (2016) warns that the law has promises and pitfalls, so a deviation is a prompt for review and not proof. Wachs et al. (2021) show that network science can describe corruption risk in contracting markets, and community detection by modularity (Newman, 2006) with the Leiden method (Traag et al., 2019) groups agencies and suppliers that award to each other.

Research Gap

The red flag studies found use data from Europe and Latin America, and the Philippine material found is about the law and the open contracting commitment (Republic Act No. 9184; Open Government Partnership, n.d.). The searches found no peer-reviewed study that combines first-digit checks, date checks and network measures into one transparent score for PhilGEPS awards, and then tests the score against simpler single indicators. This study will contribute that score and its stability tests, while clearly treating it as a screening aid. This means not found by these searches and not that none exists.

2. Problem Statement

Published procurement awards are too many to review by hand, and a single indicator such as an unusual first digit is easy to misread. The study addresses the problem of whether a transparent combination of first-digit, date and network indicators ranks awards more stably than any single indicator, and whether it helps a reviewer choose which awards to look at first.

The study will answer the following problem: How can published PhilGEPS awards be prepared and analyzed to combine first-digit, date and network indicators into a transparent risk score for review?

3. Objectives

General Objective

To develop and evaluate a transparent red flag score for Philippine government procurement awards using Benford's law checks, date checks and network analysis.

Specific Objectives

- Reproduce the counts, the missing values and the date checks on the compilation from 2021 onward, including awards dated before publication and awards with future dates, and record the source link, retrieval date and licence.

- Measure first-digit patterns against the Benford expectation for the awards, agencies and suppliers with enough awards, and report the deviations.
- Build the network of agencies and suppliers, measure its structure and detect communities.
- Combine the indicators into one transparent risk score, and test whether the rankings change when one component is removed.
- Compare the score with a single first-digit flag and with date-only, Benford-only and network-only scores, using flag rates, score stability and the overlap of the top-ranked awards.
- Compare the score with known cases only if named, verified sources can be matched to awards, and deliver a ranked review list with the reasons for each rank.

4. **Research Questions**

1. How do the first digits of award values, the award dates and the agency and supplier network behave in the published awards?
2. Does the combined score rank awards more stably than the single indicators?
3. How much do the top-ranked awards change when one component of the score is removed?
4. Can the score be checked against named, verified cases, and if not, what does that limit?

5. **Dataset and Variables**

Data item	Description
Dataset	Open compilation of PhilGEPS awards (csiiiv/philgeps-awards-dashboard), described as MIT licensed [source link, retrieval date and licence to verify]
Source	Public, published awards. The compilation reports 5,481,161 awards from 2013 to 2025 and 2,769,747 from 2021 onward [UNVERIFIED]
Unit of analysis	One published award
Target	A transparent risk score and ranking for review (unsupervised); a comparison with verified cases only if labels exist
Predictors	Award value digits, award and publication dates, agency, supplier and area identifiers, and network measures. Procurement mode and funding source are reported 100 percent empty
Available records	2,769,747 awards from 2021 onward by the group's count, to recheck. Early checks found first digit 9 in 7.8 percent of awards against 4.6 percent expected, and 5.5 percent of awards dated before publication [UNVERIFIED]

Data Preparation

- Reproduce the compilation's counts, denominators and date parsing before any result is reported.
- Exclude future-dated and impossible dates from the analysis and report how many were excluded.

- Standardize agency and supplier identifiers, and report what share of awards has a usable identifier.
- Build the network and the score components on earlier awards and evaluate on later awards, so that later links are never used to score earlier records.

6. Proposed Method

Phase	Planned work
Data acquisition	Download the compilation, record the source link, retrieval date and licence, and reproduce the counts for 2021 onward.
Exploration	Check missing values, date problems and the first-digit distribution against the Benford expectation.
Preprocessing	Parse dates, exclude impossible dates, standardize agency and supplier identifiers, and build the agency-supplier network.
Modeling	Use a single first-digit flag as the baseline, then build the combined score from the Benford, date and network indicators, with network communities found by the Leiden method.
Evaluation	Compare the combined score with the date-only, Benford-only and network-only scores, and report flag rates, score stability, component-removal tests and the overlap of the top-ranked awards.
System development	Build a ranked review list with the reason for each rank.
Interpretation	Explain each score component, and state that the score is a screening signal and not the probability of corruption.

7. Scope, Limitations, and Ethics

Scope

- The study covers published PhilGEPS awards from 2021 onward in the open compilation.
- It builds a transparent risk score and a ranked review list, and it compares the score with simpler indicators.
- A supervised model is used only if named, verified cases can be matched to awards, and otherwise the study stays unsupervised.

Limitations

- The data is open and governmental, so the study fits only if the adviser allows a title without a private client.
- Procurement mode and funding source are reported empty, so they cannot be used to explain a rank.
- An unusual indicator is a screening signal and not evidence of wrongdoing, and no award is called fraudulent.
- The counts, source link and retrieval date are [UNVERIFIED] until reproduced.

Ethical and Privacy Considerations

- Use the compilation under the terms of its source and licence, and cite where each count comes from.
- Follow the Data Privacy Act of 2012 (Republic Act No. 10173): report patterns for agencies and suppliers and avoid naming individuals.
- No human participants and no web scraping are involved. If a private watchdog, audit, compliance or civic tech group supplies records, written permission is needed and collection waits for IREC approval.
- Present results as review priorities, not as accusations.

8. Significance of the Study

The study will demonstrate a complete data science workflow on a very large public data set: data checking, exploration, indicator design, network analysis, a transparent score and a stability test. If a permitted user exists, audit and compliance teams could use the ranked list to decide which awards to review first, and civic tech groups and the public could see clearer patterns in published awards. The proposed links are SDG 16.5 and 16.6, and the study will not directly measure progress toward either goal. [The group should confirm the SDG with the adviser.]

9. Expected Output

- A documented analysis dataset with the reproduced counts and the exclusions.
- Visualizations of the first-digit, date and network patterns.
- A transparent combined risk score and a comparison with the single indicators.
- A ranked review list with the reason for each rank.
- A discussion of the limits of screening published procurement data.

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Philippine Air Cargo Route and Commodity Profiles Using K-Means Clustering

Abstract

Freight forwarders, exporters, airports and agencies plan air cargo services around which routes carry what. This study proposes a profiling system for Philippine air cargo that groups routes by value and commodity mix with K-Means clustering, adds route network analysis to show hubs and gaps, and flags unusual values with anomaly detection followed by a manual check. It uses the Philippine Statistics Authority OpenSTAT air cargo table 2L/DTR/AN/0012L4DDAA0.px, in which the group's checks found 10,099 route flows with a value from 2021 to 2025, of which 7,767 are non-food. These are aggregated route flows and not shipment records, and the source link, retrieval date and counts are [UNVERIFIED] until reproduced. A freight forwarder or cargo operator could instead supply nonpersonal shipment records as the main private dataset, with the PSA table kept as a national benchmark.

However, the studies found on air cargo are mostly about airline networks, operations and other countries, and none found profiles Philippine air cargo routes and commodity mix from PSA route flows. The recent data is thin, the 1997 values exactly copy 1995 in the group's check, and the rows are aggregated, so the study reports these limits as findings. Simple route summaries will be the baseline and K-Means the primary model, with a second clustering approach for comparison and an anomaly detector used only if the available measures support it. Profiles will be fit on earlier years and later years inspected separately, with cluster stability, the silhouette score and the rate and review outcome of anomaly flags as the evidence.

The study aims to give freight forwarders, exporters, airports and agencies a clearer view of how Philippine air cargo routes differ. However, the results describe reported route flows and not shipments, and the actual users and decisions still need to be confirmed.

1. Background and Rationale

Philippine air cargo is reported by several public sources. Reports based on data from the Civil Aeronautics Board put the air cargo handled by local airlines on domestic routes at 139.782 million kilograms in the first half of 2026, 0.9 percent more than a year earlier (PortCalls, 2026), and the Philippine Competition Commission (2020) has reviewed competition in the air transport industry, mainly the airline passenger business. The Philippine Statistics Authority publishes route-level air cargo values in its OpenSTAT database (Philippine Statistics Authority, n.d.), which is the data source of this study.

Air cargo has been studied as a network and as an operations problem. Feng et al. (2015) review air cargo operations and compare them with practice, Bombelli (2020) analyzes the topology of

the global networks of cargo integrators, Yu and Zou (2022) study air trade, air cargo demand and network analysis for the United States, and Lee et al. (2019) model air cargo network design and flight route selection. Guimerà et al. (2005) show that the worldwide air transportation network has anomalous centrality and a community structure, and communities in networks can be found by modularity (Newman, 2006) and the Leiden method (Traag et al., 2019). K-Means (Lloyd, 1982) is one of the most used clustering methods (Jain, 2010), and the quality of clusters can be checked with the silhouette score (Rousseeuw, 1987), the Davies-Bouldin index (Davies & Bouldin, 1979) and cluster stability (Hennig, 2007). Anomaly detection is surveyed by Chandola et al. (2009), and two common methods are the local outlier factor (Breunig et al., 2000) and isolation forest (Liu et al., 2008).

Research Gap

The air cargo studies found are about airline and integrator networks, operations and other countries, and the Philippine material found is about industry competition and news reports of cargo volumes (Philippine Competition Commission, 2020; PortCalls, 2026). The searches found no peer-reviewed study that profiles Philippine air cargo routes and commodity mix from PSA route flows with clustering, network analysis and anomaly checks. This study will contribute those profiles and a plain account of what the PSA table can and cannot support. This means not found by these searches and not that none exists.

2. Problem Statement

Route-level air cargo values are published, but they are not summarized into profiles that different users can compare, and the recent years of the table are thin and contain suspect values. The study addresses the problem of whether clustering, network analysis and anomaly checks can turn the PSA route flows into stable and understandable route profiles.

The study will answer the following problem: How can the Philippine air cargo route flows in the PSA OpenSTAT table be prepared and analyzed to group routes into stable profiles by value and commodity mix, to show the route network and to flag unusual values for review?

3. Objectives

General Objective

To develop and evaluate route and commodity profiles of Philippine air cargo using K-Means clustering, route network analysis and anomaly checks on PSA route flows.

Specific Objectives

- Reproduce the row counts, the time detail, the route fields and the commodity codes in the PSA table for 2021 to 2025, and record the table, the source link and the retrieval date.
- Review the data quality, including the exact match of the 1997 values with 1995 and the missing values in recent years.
- Summarize the routes with simple measures as the baseline, and cluster them with K-Means, comparing cluster counts and a second clustering approach.

- Build the route network from the route endpoints if they are usable, and find hubs and gaps and route communities.
- Test an anomaly detector only if the available measures support it, and review each flag manually before calling a value unusual.
- Report cluster stability, the silhouette score and the rate and review outcome of anomaly flags, and check whether the profiles change when the suspect years are excluded.

4. **Research Questions**

1. What route and commodity patterns appear in the PSA air cargo route flows from 2021 to 2025?
2. How many stable route profiles does K-Means find, and do they agree with a second clustering approach?
3. Which routes act as hubs or gaps in the route network?
4. Which values are flagged as unusual, and how many of them are real after a manual check?

5. **Dataset and Variables**

Data item	Description
Dataset	Philippine Statistics Authority OpenSTAT table 2L/DTR/AN/0012L4DDAA0.px, air cargo route flows (public, aggregated data)
Source	PSA OpenSTAT. The table reports 147,932 valued route flows from 1995 to 2025, of which 116,637 are non-food [UNVERIFIED]
Unit of analysis	One valued route flow (aggregated), not a shipment
Target	Route profiles (clustering, unsupervised) and anomaly flags for review
Predictors	Route endpoints, year, value and commodity category, if the table provides them in a consistent form
Available records	10,099 valued route flows from 2021 to 2025, of which 7,767 are non-food, by the group's count on 2026-10-16 (to recheck). The 1997 values exactly copy 1995

Data Preparation

- Reproduce the row counts, the route fields and the time detail before any result is reported.
- Read the commodity categories consistently, and keep food and non-food flows separate.
- Exclude or flag the suspect 1995 and 1997 values and report how the profiles change without them.
- Fit the profiles on earlier years and inspect later years separately, instead of mixing all periods before evaluation.

6. **Proposed Method**

Phase	Planned work

Data acquisition	Download the PSA table, record the table code and retrieval date, and reproduce the counts for 2021 to 2025.
Exploration	Summarize the value by route, year and commodity, and check missing values, duplicates and the 1997 copy of 1995.
Preprocessing	Read the commodity categories consistently, scale the measures and build the route network if the endpoints are usable.
Modeling	Use simple route summaries as the baseline and K-Means as the primary model, compare cluster counts and a second clustering approach, and test an anomaly detector only if the measures support it.
Evaluation	Report cluster stability, the silhouette score, and the rate and manual review outcome of anomaly flags.
System development	Build a route profile view that shows each cluster, the hubs and gaps and the flagged values.
Interpretation	Describe each profile in plain words, and check whether the profiles change when the suspect years are excluded.

7. Scope, Limitations, and Ethics

Scope

- The study covers the PSA OpenSTAT air cargo route flows from 2021 to 2025, with earlier years used only for data quality checks.
- It profiles routes by value and commodity mix, shows the route network and flags unusual values for review.
- Shipment-level validation is possible only if a freight forwarder or cargo operator supplies nonpersonal records.

Limitations

- The data is open and aggregated, so the study fits only if the adviser allows a title without a private client and accepts route flows as rows.
- The 7,767 recent non-food flows pass the 5,000-row rule but not a 10,000-row target.
- The table reports route flows and not shipments, so the study cannot claim airport-level or monthly results unless the table provides them.
- The counts, table link and retrieval date are [UNVERIFIED] until reproduced.

Ethical and Privacy Considerations

- Use the PSA data under the terms of its source and cite the table and the retrieval date.
- No personal data, no human participants and no web scraping are involved in the PSA data.
- Follow the Data Privacy Act of 2012 (Republic Act No. 10173): if a freight forwarder or cargo operator supplies records, they must be nonpersonal, supplied with the written permission of an authorized representative, and collected only after IREC approval.
- Present flagged values as items for review and not as errors or wrongdoing.

8. **Significance of the Study**

The study will demonstrate a complete data science workflow on public trade data: data checking, exploration, clustering, network analysis, anomaly detection and a clear account of what the data can support. Freight forwarders and exporters could compare route profiles when planning services, and airports and agencies could use a national view of reported flows. These uses are proposed and the actual decisions need to be confirmed with users. The link to a Sustainable Development Goal is for the adviser to confirm.

9. **Expected Output**

- A documented analysis dataset with the reproduced counts and the quality checks.
- Visualizations of the route and commodity patterns.
- Route profiles with cluster stability and the silhouette score.
- A route network view with hubs, gaps and the flagged values.
- A discussion of the limits of profiling aggregated route flows.

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